Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Schrödinger equation is equivalent to information conservation

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy.lean`
- Library: `UnitaryEntropy`
- Imports: `Mathlib.Analysis.SpecialFunctions.BinaryEntropy`, the Wigner library files `EqualMixtures.lean`, `Kadison.lean` and `Differentiable.lean` (which import the rest of `ForPhyslib/QuantumMechanics/Wigner`), and two new `ForMathlib` files:
  `ForMathlib/Analysis/Calculus/Deriv/Star.lean` (in a normed star algebra, the derivative of a path of unitaries through 1 is skew-adjoint, and $U^\dagger U$ is constant when $U' = XU$ with X skew-adjoint),
  `ForMathlib/Analysis/InnerProductSpace/Adjoint.lean` (in finite dimension, $U^\dagger U = 1$ already makes U unitary)
- Wigner library, in `ForPhyslib/QuantumMechanics/Wigner/`: `Density` (density operators, pure states `proj`, equal mixtures `mix`; moved there from this brief), `Basic`, `Normalize`, `Rigidity`, `Theorem` (Wigner's theorem without bijectivity, after Gehér, now in any

dimension, see Section 1.2; uniqueness up to one phase in finite dimension), `Projection` (the same for maps on rank-one projections), `Continuity` (Bargmann's triple product excludes the anti-unitary case near t_0), `Differentiable` (explicit, differentiable choice of the phase). Its `ForMathlib` helpers are `Analysis/Complex/ThreeCircles.lean`, `Analysis/InnerProductSpace/PhaseUnitary.lean` and `Analysis/InnerProductSpace/LinearMap.lean` (conjugate-linear isometries conjugate the inner product; polarization of $|x\rangle\langle y|$). Coordinate conjugation in an orthonormal basis came from TauCeti (`TauCeti.Analysis.InnerProductSpace.Conjugation`) until the library moved to Hilbert bases.

Setting. E is a finite-dimensional complex Hilbert space, and operators are continuous linear maps $E \to E$. A density operator (`IsDensity`) is a positive operator of trace 1. The pure state of a unit vector ψ is `proj` $\psi = |\psi\rangle\langle\psi|$ (Mathlib's `rankOne`), and `mix` $\rho\,\sigma = \frac{1}{2}\rho + \frac{1}{2}\sigma$. The von Neumann entropy `vonNeumann` is $\sum -\mu \log \mu$ over the roots μ of the characteristic polynomial. For self-adjoint ρ these are its eigenvalues (`vonNeumann_eq_sum`), and the definition makes invariance under conjugation immediate. `born` $\psi\,\phi = |\langle\psi|\phi\rangle|^2/(\|\psi\|^2\|\phi\|^2)$, and `mixEntropy` $p = H_2((1+\sqrt{p})/2)$ with H_2 Mathlib's `binEntropy`. The Schrödinger equation for a propagator is $U'(t) = (i\hbar)^{-1}H(t)\,U(t)$ with $H(t)$ self-adjoint.

Results.

Brief	Lean
Corollary 1, entropy of $\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$	`vonNeumann_mix_proj`, `mixEntropy_eq`
Corollary 1, strictly decreasing	`mixEntropy_strictAntiOn` (on $[0,1]$)
Corollary 1, $p(\phi\|\psi) = \lambda$ iff $S = \ldots$	`born_eq_iff` (for $\lambda \in [0,1]$)
Theorem 2 ($\Rightarrow$), unitary propagator	`star_mul_self_of_schrodinger`, `ContinuousLinearMap.mem_unitary_of_star_mul_self`
Theorem 2 ($\Rightarrow$), existence for constant H	`hasDerivAt_exp_schrodinger`: $U(t) = \exp(t(i\hbar)^{-1}H)$
Theorem 2 ($\Rightarrow$), existence for continuous $H(t)$	`exists_propagator_of_continuous` (Section 1.3)
Theorem 2 ($\Rightarrow$), DR-INFO	`vonNeumann_unitary_conj`, `IsDensity.unitary_conj`, `schrodinger_conserves_entropy`
Theorem 2 ($\Leftarrow$), Step 1	`vonNeumann_eq_zero_iff`: $S(\rho) = 0$ iff ρ is pure
Theorem 2 ($\Leftarrow$), Steps 1–2 up to Wigner	`pure_and_born_of_preserves`, `proj_and_trace_of_preserves`
Theorem 2 ($\Leftarrow$), Step 2, Wigner's theorem	`Wigner.wigner_rankOne` (from `Wigner.wigner`)
Theorem 2 ($\Leftarrow$), anti-unitary case excluded	`Wigner.eventually_exists_unitary`, `PhaseEqOnSphere.false_of_unitary_of_antiunitary`
Theorem 2 ($\Leftarrow$), differentiable $U(t)$	`Wigner.exists_unitary_differentiableAt`
Theorem 2 ($\Leftarrow$), Step 3	`exists_hamiltonian` (from `star_eq_neg_of_hasDerivAt_of_mem_unitary`), for $\hbar \neq 0$
Theorem 2 ($\Leftarrow$), mixed states	`Wigner.affineOnDensities_of_map_mix`, `Wigner.AffineOnDensities.eq_conjOp`, `Wigner.mul_mul_star_eq_conjOp`
Theorem 2 ($\Leftarrow$), complete	`schrodinger_of_preserves`

Differences from the brief.

- This part is finite-dimensional; the brief has a separable Hilbert space. The calculus results (`star_mul_self_of_schrodinger`, `exists_hamiltonian`) hold in any Hilbert space for bounded $H(t)$. Corollary 1 and Theorem 2 through Wigner's theorem are redone in any dimension in Section 1.2.

- Corollary 1 is proved without diagonalizing the mixture. $\rho = \frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$ satisfies $\rho^3 = \rho^2 - \frac{1-\lambda}{4}\rho$ (`mix_proj_cubic`), so every eigenvalue is 0 or $(1 \pm \sqrt{\lambda})/2$. On these three points $-\mu \log \mu = \alpha\mu + \beta\mu^2$ for suitable α, β, and $\operatorname{tr}\rho = 1$, $\operatorname{tr}\rho^2 = (1+\lambda)/2$ fix the sum. This also covers the degenerate cases $\lambda = 0$ (double eigenvalue $\frac{1}{2}$) and $\lambda = 1$ (ρ pure).

B EVMIX is used only for equal mixtures, which is all Step 2 needs. For the ($\Rightarrow$) direction, $\rho \mapsto U\rho U^\dagger$ is linear, so it preserves all mixtures.

- DR-INFO is $S(\rho_I) = S(\rho_F) = S(\rho_P)$. The reconstructed past $U^\dagger \rho_F U$ is ρ_I itself, which is also proved.

- The ($\Rightarrow$) direction shows that every propagator solving the equation is unitary. Existence is shown for constant H by the exponential, and for a continuous family of bounded $H(t)$ by the global existence theorem for linear ODEs, which the brief assumes (Section 1.3).

- `schrodinger_of_preserves` assumes $\Phi_{t_0} = \mathrm{id}$ and differentiability of $t \mapsto \Phi_t(\rho)$ at t_0 only on density operators, one ρ at a time. It concludes that $\Phi_t(|\psi\rangle\langle\psi|) = |U(t)\psi\rangle\langle U(t)\psi|$, and $\Phi_t(\rho) = U(t)\rho U(t)^\dagger$ for every density operator ρ, for t near t_0, with $U(t)$ unitary for every t, $U(t_0) = 1$ and $U'(t_0) = (i\hbar)^{-1}HU(t_0)$ for a self-adjoint H. This is the Schrödinger equation at t_0, which is what EVDIFF at t_0 gives. For a two-parameter evolution, one propagator solves it at every time (Section 1.1).

- Wigner's theorem is used in its non-bijective form (Gehér), so the bijectivity of the correspondence of pure states via the reconstructed past (Step 1) is not needed.

- $\Phi_t(\rho) = U(t)\rho U(t)^\dagger$ for mixed ρ follows from the pure states once Φ_t is affine, by the spectral decomposition of ρ (`Wigner.AffineOnDensities.eq_conjOp`). Affinity is not an extra assumption: a map of the density operators that preserves equal mixtures preserves all convex combinations of two of them (`Wigner.affineOnDensities_of_map_mix`), since it is bounded and preserves midpoints.

Problems found in the brief. No errors. Implicit points:

- The "iff" in Corollary 1 needs $\lambda \in [0, 1]$, where the entropy is strictly decreasing in λ; for other λ, $\sqrt{\lambda}$ and the logarithms are not meaningful.

- Step 3 assumes a norm-differentiable unitary family, which forces a bounded generator. On an infinite-dimensional Hilbert space it therefore covers bounded Hamiltonians only, and even then EVDIFF does not give such a family (Section 1.4). The brief "Multiple equivalent characterizations of the Schrödinger equation" treats the general case with Stone's theorem.

- In Step 3, $\hbar \neq 0$ is needed to write $A = H/(i\hbar)$.

- Wigner's theorem gives each $U(t)$ only up to a phase, and Step 3 differentiates $U(t)$. A differentiable choice exists. Extend Φ_t to all $|x\rangle\langle y|$ by polarization, so that it sends $|x\rangle\langle y|$ to $|Ux\rangle\langle Uy|$, and fix an orthonormal basis (b_k) and an index i. Then

$$U(t) = \left(\mathrm{Re}\,\langle b_i|\Phi_t(|b_i\rangle\langle b_i|)\,b_i\rangle\right)^{-1/2} \sum_k \Phi_t(|b_k\rangle\langle b_i|)\ |b_i\rangle\langle b_k|$$

  is the unitary with $\langle b_i|U(t)b_i\rangle > 0$. It is a finite expression in values of Φ_t on pure states, so it is differentiable at t_0 whenever these are.

- "Joint continuity and $\Phi_{t_0} = I$ exclude the anti-unitary case" needs only continuity at t_0 of $t \mapsto \Phi_t(P)$ for three pure states P: the imaginary part of Bargmann's triple product $\mathrm{tr}(\Phi_t(P_1)\Phi_t(P_2)\Phi_t(P_3))$ is $\frac{1}{4}$ at t_0 and would be $-\frac{1}{4}$ for an anti-unitary. In dimension 1 there is nothing to exclude.

1.1 The equation at every time

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/TwoParameter.lean`

- Imports: `SchroedingerEvolution.TwoParameter`, from the brief "Multiple equivalent characterizations of the Schrödinger equation" (for the two-parameter `IsEvolution` and `DrSceq` with $H(t)$; it imports this brief's main file), and the new `ForPhyslib/QuantumMechanics/Wigner/Propagator.lean` (sections of determinant one, their uniqueness up to a root of unity, and gluing along an interval).
- New `ForMathlib` file `ForMathlib/Analysis/Calculus/ContDiff/FiniteDimension.lean`: the determinant is a smooth function of the operator (Mathlib has only its continuity). `ForMathlib/Analysis/Calculus/Deriv/Star.lean` gains `star_mul_star_eq_neg_of_hasDerivAt_of_mem_unitary`: for a path of unitaries, $U'(t)U(t)^\dagger$ is skew-adjoint at every t. `Wigner.exists_unitary_differentiableAt` now also gives differentiability at every t near t_0 at which the family is differentiable.

Setting. E is finite-dimensional. A two-parameter evolution $\Phi_{t_0\to t_1}$, $t_0 \le t_1$ (`IsEvolution`), maps density operators affinely to density operators, with $\Phi_{t\to t} = I$, $\Phi_{t_1\to t_2}\circ\Phi_{t_0\to t_1} = \Phi_{t_0\to t_2}$, and $(\rho, t_0, t_1) \mapsto \Phi_{t_0\to t_1}(\rho)$ jointly continuous. There is no EVTI. DR-INFO (`DrInfo`) is $S(\Phi_{t_0\to t_1}(\rho)) = S(\rho)$ for all $t_0 \le t_1$. EVDIFF (`EvDiff`) is: for every reference time r and density operator ρ, $t \mapsto \Phi_{r\to t}(\rho)$ is differentiable at every $t > r$. DR-SCEQ (`DrSceq`) is conjugation, $\Phi_{t_0\to t_1}(\rho) = U(t_1)U(t_0)^\dagger\,\rho\,(U(t_1)U(t_0)^\dagger)^\dagger$, by a U with $U(0) = 1$ and $U'(t) = (i\hbar)^{-1}H(t)U(t)$ at every t, for self-adjoint $H(t)$.

Results.

Brief	Lean
DR-INFO, EVDIFF for $\Phi_{t_0\to t_1}$	`DrInfo`, `EvDiff`
Each $\Phi_{r\to t}$ is unitary	`exists_unitary`
Local unitaries of determinant one	`Wigner.eventually_isDetOneSection`
Uniqueness up to a constant root of unity	`Wigner.IsDetOneSection.exists_eq_smul`, `Wigner.eq_inner_smul_of_proj_eq`
One differentiable $U(t)$ on an interval	`Wigner.exists_isDetOneSection` (gluing: `IsPreconnected.exists_forall_of_eventually`), `exists_section`
One differentiable $U(t)$ for all t	`exists_propagator`
Step 3 at every time	`star_mul_star_eq_neg_of_hasDerivAt_of_mem_unitary`, `exists_hamiltonian_of_star_eq_neg`
Theorem 2, at every time	`drSceq_iff_drInfo_and_evDiff`

Differences from the brief.

- Finite dimension only; see the next paragraph for why the argument does not extend.
- EVDIFF asks for differentiability of $t \mapsto \Phi_{r\to t}(\rho)$ only at $t > r$, two-sided. There is no differentiability in the initial time and no one-sided derivative at $t = r$; a reference time $r < \tau$ gives the two-sided derivative of U at τ. A jump $H(t) = H_\beta$ for $t < 0$, $H(t) = H_\alpha$ for $t \ge 0$, with $H_\alpha - H_\beta$ not a multiple of 1, violates EVDIFF at $t = 0$ (the one-sided derivatives $(i\hbar)^{-1}[H_{\alpha,\beta}, \rho(0)]$ differ for some ρ), as it must: the U of DR-SCEQ is differentiable at every t.
- DR-INFO has no reconstructed past; in finite dimension it is not needed.

- The proof of ($\Leftarrow$). By Wigner's theorem each $\Phi_{r\to t}$ is induced by a unitary or an anti-unitary. Near any t' the maps $\Phi_{\min(t,t')\to\max(t,t')}$ are close to the identity and so unitary (`Wigner.eventually_exists_unitary`), and by the composition law $\Phi_{r\to t}$ has the type of $\Phi_{r\to t'}$; the type is locally constant, and $\Phi_{r\to r} = I$. Near $\tau > r$, conjugating by a unitary that induces $\Phi_{r\to\tau}$ makes the family the identity at τ, and the explicit `Wigner.unitaryOf` is differentiable near τ; multiplying by a branch of $\det(\cdot)^{-1/n}$, $n = \dim E$, gives determinant one. Two such families that induce the same maps differ by $c(t) = \langle W(t)e|V(t)e\rangle$, which is continuous with $c(t)^n = 1$, hence constant on a connected set. So the local families glue along (r,∞): the set of times reached from a fixed time is open and closed in (r,∞). The families for $r = -1, -2, \ldots$, normalized by $V(0)^\dagger$, agree, and give U; then $H(t) = i\hbar U'(t)U(t)^\dagger$.
- The propagator $U_{t_0\to t_1} = U(t_1)U(t_0)^\dagger$ satisfies the composition law exactly; no phase cocycle appears.

Problems found in the brief.

- The proof of Theorem 2 differentiates at one time, with phases chosen near that time. Going from "at each time" to one differentiable propagator for all times needs the local phase choices to be glued, and locally they are fixed only up to a continuous phase function. Fixing the determinant reduces this freedom to an n-th root of unity, which connectedness removes. The determinant needs finite dimension, so the infinite-dimensional case stays open.

1.2 Hilbert spaces of any dimension

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/InfiniteDimensional.lean`
- Library: `UnitaryEntropy`, which now builds every module `UnitaryEntropy.*`
- Imports:
  `ForPhyslib.QuantumMechanics.VonNeumannEntropy`,
  `ForPhyslib.QuantumMechanics.EqualMixtures` and `ForPhyslib.QuantumMechanics.Kadison` (equal mixtures give affinity, and the mixed-state extension),
  `ForMathlib.Analysis.InnerProductSpace.StrongDerivative`,
  `TauCeti.Analysis.InnerProductSpace.FunctionalCalculusComparison` (eigenvector action of the functional calculus, for the comparison in finite dimension),
  and the finite-dimensional part `UnitaryEntropy` (for `born`, `mixEntropy`, `mix_proj_cubic`, `mix_proj_sq`, `star_mul_self_of_schrodinger` and `exists_hamiltonian`, none of which assumes finite dimension)
- New file `ForPhyslib/QuantumMechanics/DensityOperator.lean`: density operators on a Hilbert space of any dimension, on top of the trace-class operators of PhyslibAlpha (`PhyslibAlpha/ProbabilisticTheory/HilbertSpace/TraceClass/`, from the physlib dependency; it builds against our Mathlib pin). Pure states, convex combinations, conjugation by an isometry, idempotent density operators are pure, and agreement with `IsDensity` in finite dimension.
- New file `ForPhyslib/QuantumMechanics/VonNeumannEntropy.lean`: the von Neumann entropy in $[0,\infty]$, its invariance under conjugation by an isometry, and $S(\rho) = 0$ iff ρ is pure.

- New `ForMathlib` material:
  `ForMathlib/Analysis/InnerProductSpace/PositiveTrace.lean` (the trace of a positive operator in $[0,\infty]$, independent of the Hilbert basis),
  `ForMathlib/Analysis/InnerProductSpace/HilbertBasisMaps.lean` (for a Hilbert basis: a vector is determined by its coordinates, Parseval for their moduli, the diagonal unitary $b_i \mapsto \delta_i b_i$, coordinatewise conjugation, the isometry $b_i \mapsto v_i$ onto an orthonormal family, and $V(V^\dagger x) = x$ when $\|V^\dagger x\| = \|x\|$),
  `ForMathlib/Analysis/InnerProductSpace/StrongDerivative.lean`
  (by Banach–Steinhaus, the strong derivative of an operator family is a bounded operator; for unitaries with $U(t_0) = 1$ it is skew-adjoint).

- The Wigner library `ForPhyslib/QuantumMechanics/Wigner/` (`Basic`, `Normalize`, `Rigidity`, `Theorem`) now works over a Hilbert basis of any Hilbert space. `Theorem` proves Wigner's theorem in any dimension, with isometries, and derives the finite-dimensional theorem from it. `Continuity` excludes the anti-unitary branch in any dimension, without a trace, and derives the finite-dimensional exclusion from it. `Differentiable` adds the differentiable choice of the phase in any dimension. The other files built on the library are unchanged.

Setting. E is a complex Hilbert space of any dimension; separability is not used. `posTrace` $T = \sum_i \langle e_i | T e_i \rangle \in [0,\infty]$ for a Hilbert basis (e_i). For positive T it does not depend on the basis (`posTrace_eq_tsum`): write $\langle e_i | R R^\dagger e_i \rangle = \|R^\dagger e_i\|^2$, expand by Parseval in a second basis and exchange the two sums of nonnegative terms. A density operator (`IsDensityOperator`) is a positive operator that is trace class in the sense of PhyslibAlpha (`IsTraceClass`) with `trace` $\rho = 1$; equivalently, $0 \le \rho$ and `posTrace` $\rho = 1$ (`isDensityOperator_iff`). The entropy is `vonNeumannEntropy` $\rho =$ `posTrace` $\eta(\rho)$ with $\eta(x) = -x \log x$ and $\eta(\rho)$ given by the continuous functional calculus. The spectrum of a density operator lies in $[0,1]$, where $\eta \ge 0$, so $\eta(\rho)$ is positive. `proj`, `mix`, `born` and `mixEntropy` are as above.

Results.

Brief	Lean
QST:H, density operators	`IsDensityOperator`, `isDensityOperator_proj`, `IsDensityOperator.mix`
QST:H, von Neumann entropy	`vonNeumannEntropy`
Agreement in finite dimension	`isDensityOperator_iff_isDensity`, `vonNeumannEntropy_eq_ofReal_vonNeumann`
Corollary 1, entropy of $\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$	`vonNeumannEntropy_mix_proj`
Corollary 1, $p(\phi\|\psi) = \lambda$ iff $S = \dots$	`born_eq_iff_vonNeumannEntropy` (for $\lambda \in [0,1]$)
Theorem 2 ($\Rightarrow$), unitary conjugation preserves the entropy	`IsDensityOperator.vonNeumannEntropy_conj` (for isometries)
Theorem 2 ($\Rightarrow$), DR-INFO	`IsDensityOperator.conj`, `schrodinger_conserves_vonNeumannEntropy`
Theorem 2 ($\Leftarrow$), Step 1	`IsDensityOperator.vonNeumannEntropy_eq_zero_iff`
Theorem 2 ($\Leftarrow$), Steps 1–2 up to Wigner	`pure_and_born_of_preserves_vonNeumannEntropy`, `exists_isWignerMap_of_preserves`
Wigner's theorem, any dimension	`Wigner.IsWignerMap.exists_linearIsometry_or_conjLinearIsometry`, `Wigner.wigner_isometry`
Wigner's theorem, onto the rays	`Wigner.IsWignerMap.exists_unitary_or_antiunitary_of_surjective`
Theorem 2 ($\Leftarrow$), Steps 1–2 with Wigner	`exists_isometry_of_preserves`; `exists_unitary_of_preserves` if every pure state is an image
Theorem 2 ($\Leftarrow$), surjectivity from the reconstructed past	`exists_proj_of_past`, `exists_unitary_of_preserves_of_past`
Theorem 2 ($\Leftarrow$), anti-unitary case excluded	`eventually_exists_unitary_of_preserves` (from `Wigner.eventually_exists_unitary_of_unitary_or_antiunitary`)
Theorem 2 ($\Leftarrow$), surjectivity is needed	`shift_preserves_not_unitary`
Theorem 2 ($\Leftarrow$), differentiable $U(t)\psi$	`Wigner.exists_unitary_differentiableAt_apply`
Theorem 2 ($\Leftarrow$), Step 3	`ContinuousLinearMap.exists_hasDerivAt_apply`, `star_eq_neg_of_hasDerivAt_apply_of_mem_unitary`, `exists_hamiltonian_of_star_eq_neg`
Theorem 2 ($\Leftarrow$), mixed states	`affineOnDensityOperators_of_map_mix`, `AffineOnDensityOperators.eq_conjOp`
Theorem 2 ($\Leftarrow$), complete	`schrodinger_of_preserves_vonNeumannEntropy`

Differences from the brief.

- Wigner's theorem in any dimension follows Gehér's proof with a Hilbert basis (b_k). The first normalization changes: the images $f(b_k)$ are orthonormal but need not span, so they define an isometry V with $Vb_k = f(b_k)$, not a unitary. By Parseval, $\|V^\dagger f(v)\|^2 = \sum_k |\langle f(b_k)|f(v)\rangle|^2 = \sum_k |\langle b_k|v\rangle|^2 = 1 = \|f(v)\|^2$, so $f(v)$ lies in the range of V and $V^\dagger \circ f$ is a Wigner map that fixes every b_k. The diagonal unitary and the conjugation are then built from the basis as in finite dimension. The result is a linear or conjugate-linear isometry; it is onto, hence unitary or anti-unitary, when every ray is the image of a ray, and always in finite dimension.
- The anti-unitary branch is excluded by continuity without a trace. For $P = |a\rangle\langle a|$, Q,

R rank-one projections onto a, b, c and a fixed vector x, $\langle x|PQRP\,x\rangle = |\langle a|x\rangle|^2\,\Delta(a,b,c)$ (`inner_rankOne_mul_rankOne_mul_rankOne_mul_rankOne_apply`). Take the test triple $b_i, (b_i - b_k)/\sqrt{2}, (b_i - ib_k)/\sqrt{2}$ with $\Delta = (1+i)/4$, $x = b_i$, and the images $\Phi_t(P)$. The imaginary part is $\frac{1}{4}$ at t_0 and continuous at t_0 in operator norm, but ≤ 0 for an anti-unitary, since its triple product is $\overline{\Delta}$ and the factor $|\langle a|x\rangle|^2$ is nonnegative. The finite-dimensional `Wigner.eventually_exists_unitary`, which used $\mathrm{tr}(PQR) = \Delta$, is now a corollary. Continuity is assumed in operator norm; continuity in trace norm implies it.

- EVDIFF is formalized as in finite dimension: $t \mapsto \Phi_t(\rho)$ is differentiable at t_0 in operator norm for every density operator ρ. The conclusion is the Schrödinger equation on vectors, as in DR-SCEQ: $\psi(t) = U(t)\psi$ satisfies $\psi'(t_0) = (i\hbar)^{-1}H\psi(t_0)$ for every ψ, with one bounded self-adjoint H. It is not the equation for $t \mapsto U(t)$ in operator norm, which the finite-dimensional `schrodinger_of_preserves` gives; no choice of phases gives it (Section 1.4). The basis sum that fixes the phase in finite dimension does not converge in operator norm; instead, for a fixed unit vector e, Φ_t extended by polarization gives $(|Ux\rangle\langle Ue|)\,e = \langle Ue|e\rangle\,Ux$, and dividing by $|\langle Ue|e\rangle|$ fixes the phase one vector at a time. So $t \mapsto U(t)x$ is differentiable at t_0 for each x. By Banach–Steinhaus, applied to the difference quotients, the derivatives $x \mapsto (U(\cdot)x)'(t_0)$ form a bounded operator A. Differentiating $\langle U(t)x|U(t)y\rangle = \langle x|y\rangle$ gives $A^\dagger = -A$, and $H = i\hbar A$.

- Mixed states follow as in finite dimension, with the spectral sum replaced by the canonical expansion $\rho = \sum_i \mu_i\,|e_i\rangle\langle e_i|$, which may be infinite. Write $\rho = m\rho_F + (1-m)\sigma_F$ with ρ_F a finite part of weight m; Φ_t and $U(t)\cdot U(t)^\dagger$ agree on ρ_F by affinity, so they differ on ρ by at most $2(1-m)$ in operator norm, and $m \to 1$ (`AffineOnDensityOperators.eq_conjOp`). Affinity again comes from equal mixtures (`affineOnDensityOperators_of_map_mix`).

- No spectral decomposition is used. The entropy is the trace of $\eta(\rho)$, and pure states are found through the spectrum: $S(\rho) = 0$ forces $\eta(\rho) = 0$, so η vanishes on the spectrum, which then lies in $\{0,1\}$; ρ is idempotent, and an idempotent density operator is $|\psi\rangle\langle\psi|$ (`IsDensityOperator.eq_proj_of_isIdempotentElem`).

- Corollary 1 uses the cubic relation as in finite dimension, with the spectral mapping theorem for polynomials in place of eigenvalues. Then $\eta(\rho) = \alpha\rho + \beta\rho^2$, and its trace is computed from $\sum_i \langle e_i|x\rangle\langle y|e_i\rangle = \langle y|x\rangle$ for the rank-one terms.

- The ($\Rightarrow$) direction uses only $U(t)^\dagger U(t) = 1$, which every propagator satisfies. Conjugation $X \mapsto UXU^\dagger$ by an isometry is a non-unital star algebra homomorphism, so it commutes with the functional calculus of η (since $\eta(0) = 0$), and it preserves `posTrace`. Unitarity is not needed.

- Entropies are compared in $[0, \infty]$; in infinite dimension $S(\rho) = \infty$ occurs.

- DR-INFO is formalized in full: $S(\Phi\rho) = S(\rho)$, and every density operator ρ has a reconstructed past, a density operator σ with $\Phi\sigma = \rho$; then $S(\sigma) = S(\rho)$ by the first condition. The finite-dimensional part does not need the past, and `exists_isometry_of_preserves` shows what remains without it.

Problems found in the brief.

- In infinite dimension the reconstructed past in DR-INFO is essential, and Step 1 uses it when it says that pure states "correspond bijectively". DR-INFO presupposes a past ensemble ρ_P for every initial ensemble, so Φ is onto the density operators, as in the introduction's

"deterministic and reversible". Then every pure state is the image of a pure state, since its past has entropy zero (`exists_proj_of_past`), and Wigner's theorem gives a unitary or an anti-unitary (`exists_unitary_of_preserves_of_past`). The brief's proof is therefore correct, but the past cannot be dropped. Conjugation $\rho \mapsto S\rho S^\dagger$ by the shift S on $\ell^2(\mathbb{N})$ maps density operators to density operators and preserves the entropy and equal mixtures, but no pure state goes to $|e_0\rangle\langle e_0|$. So it is induced by no unitary and by no anti-unitary (`shift_preserves_not_unitary`), and Wigner's theorem gives only an isometry. In finite dimension the past is automatic.

- The time structure of [B]EVMIX does not replace the past. On $L^2(0,\infty)$ the right shifts $(V_t f)(x) = f(x-t)$ for $x > t$ and 0 otherwise are isometries forming a strongly continuous semigroup with $V_0 = I$. Then $\Phi_{t_0\to t_1}(\rho) = V_{t_1-t_0}\rho V_{t_1-t_0}^\dagger$ satisfies [B]EVMIX, preserves the entropy, and is induced by no unitary for $t_1 > t_0$; it has no reconstructed past. (By the Wold–Cooper decomposition, every strongly continuous semigroup of isometries is a unitary group plus copies of this one.) The example is formalized as `shift_semigroup_counterexample` in the brief "Multiple equivalent characterizations of the Schrödinger equation"; the Wold–Cooper decomposition is not. A norm-continuous semigroup of isometries is $V_t = e^{tA}$ with bounded A, and differentiating $V_t^\dagger V_t = I$ at 0 gives $A^\dagger = -A$, so it is unitary. The example therefore needs an unbounded generator, which Step 3 does not cover anyway.

- EVDIFF for every density operator forces a bounded Hamiltonian, as the Banach–Steinhaus step shows. Unbounded Hamiltonians, whose evolution is differentiable only on a dense set of states, are outside the theorem as stated; see Section 1.5. Otherwise no problems beyond those of the finite-dimensional part.

1.3 The propagator of a time-dependent Hamiltonian

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/Propagator.lean`
- Imports: `UnitaryEntropy.InfiniteDimensional` (for `schrodinger_conserves_vonNeumannEntropy`), `UnitaryEntropy.TwoParameter` (for the two-parameter `IsEvolution`, `DrSceq`, `DrInfo`, `EvDiff` and `drSceq_iff_drInfo_and_evDiff`), and the new `ForMathlib.Analysis.ODE.Linear`.
- New `ForMathlib` file `ForMathlib/Analysis/ODE/Linear.lean`: global existence and uniqueness for $x' = L(t)x$ with L a continuous family of bounded operators on a Banach space (`exists_hasDerivAt_clm_apply`, `eq_of_hasDerivAt_clm_apply`), the same for $U' = A(t)U$ in a Banach algebra (`exists_hasDerivAt_mul`, `eq_of_hasDerivAt_mul`), and unitarity of the solution for skew-adjoint $A(t)$ (`mul_star_self_eq_one_of_hasDerivAt`, `exists_hasDerivAt_mul_mem_unitary`). It imports Mathlib's `Mathlib.Analysis.ODE.ExistUnique` (Picard–Lindelöf, uniqueness, Grönwall) and `ForMathlib/Analysis/Calculus/Deriv/Star.lean`.

Setting. E is a complex Hilbert space of any dimension, and $H : \mathbb{R} \to \mathcal{B}(E)$ is continuous in operator norm with every $H(t)$ self-adjoint. The brief's Theorem 2 ($\Rightarrow$) starts from "The Schrödinger equation gives rise to a differentiable unitary propagator $U_{t_0\to t_1}$". Here the propagator is constructed.

Results.

Brief	Lean
Linear ODE, global existence and uniqueness	`exists_hasDerivAt_clm_apply`, `eq_of_hasDerivAt_clm_apply`
Propagator in a Banach algebra; unitary for skew-adjoint $A(t)$	`exists_hasDerivAt_mul`, `exists_hasDerivAt_mul_mem_unitary`
Theorem 2 ($\Rightarrow$), the propagator exists and is unitary	`exists_propagator_of_continuous`
Theorem 2 ($\Rightarrow$), the propagator is unique	`propagator_unique`
Theorem 2 ($\Rightarrow$), DR-SCEQ $\Rightarrow$ DR-INFO with this propagator	`exists_propagator_conserves_vonNeumannEntropy`
Two-parameter evolution of a propagator	`TwoParameter.conjEvolution`, `TwoParameter.isEvolution_conjEvolution`
Theorem 2 ($\Rightarrow$), two-parameter, finite dimension: QST:H + [B]EVMIX, DR-SCEQ, DR-INFO, EVDIFF	`TwoParameter.exists_propagator_isEvolution`

Proof.

- Mathlib's Picard–Lindelöf theorem solves $x' = f(t, x)$ on $[t_0 - T, t_0 + T]$ only when $\|f\| \le L$ on a ball of radius a with $LT \le a$. For the linear field $f(t, x) = L(t)x$, L grows with a, so this gives only $T < 1/\sup\|L(t)\|$. On $[t_0 - T, t_0 + T]$ the equation is therefore solved for the truncated field $L(t)\pi_R(x)$, where $\pi_R(x) = \min(1, R/\|x\|)\, x$ is the radial retraction onto the ball of radius R. It is bounded by MR, with $M = \sup_{[t_0-T,t_0+T]} \|L(t)\|$, and globally 2-Lipschitz, so Picard–Lindelöf applies for every T. Since $\|\pi_R(x)\| \le \|x\|$, Grönwall's inequality on both sides of t_0 (the left side by reflection $t \mapsto 2t_0 - t$) gives $\|x(t)\| \le \|x_0\| e^{MT} < R$ for $R = \|x_0\| e^{MT} + 1$. So $\pi_R(x(t)) = x(t)$ and x solves the linear equation.
- The interval solutions agree on common intervals by Mathlib's uniqueness theorem (`ODE_solution_unique_of_mem_Icc`, Lipschitz constant M), and $x(t) = x_{|t-t_0|+1}(t)$ is a global solution: near each t it agrees with one interval solution, which is differentiable at the interior point t.
- The propagator is the case $L(t)U = A(t)U$, $A(t) = (i\hbar)^{-1}H(t)$, in the Banach algebra $\mathcal{B}(E)$. $U^\dagger U$ is constant (`star_mul_self_eq_of_hasDerivAt`), hence 1. For $UU^\dagger$, which has no such conservation law, $D = UU^\dagger - 1$ solves the linear equation $D' = AD - DA$, since $A^\dagger = -A$, with $D(t_0) = 0$; by uniqueness $D = 0$. So $U(t)$ is unitary in any dimension, not only an isometry.
- For the two-parameter evolution $\Psi_{t_0 \to t_1}(\rho) = W\rho W^\dagger$, $W = U(t_1)U(t_0)^\dagger$, the composition law is $U(t_2)U(t_1)^\dagger U(t_1)U(t_0)^\dagger = U(t_2)U(t_0)^\dagger$, and joint continuity follows from the continuity of U. DR-SCEQ then holds by definition, and DR-INFO and EVDIFF follow from `TwoParameter.drSceq_iff_drInfo_and_evDiff`.

Differences from the brief.

- $H(t)$ is assumed bounded and continuous in operator norm. For unbounded $H(t)$ the propagator needs Kato-type hypotheses and is not constructed (open problems). A family of bounded $H(t)$ that is only strongly continuous also has a strongly differentiable propagator; this is not formalized.

Problems found in the brief. None: for bounded continuous $H(t)$ the assumed propagator exists, is unique and is unitary.

1.4 No norm-differentiable propagator

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/NormDifferentiable.lean`
- Imports: `ForPhyslib.QuantumMechanics.VonNeumannEntropy` and `ForPhyslib.QuantumMechanics.Kadison` (density operators, entropy, products of rank-one projections) and `ForPhyslib.QuantumMechanics.Wigner.Propagator` (two unitaries with the same action on pure states differ by a phase, `Wigner.eq_inner_smul_of_proj_eq`).

Setting. In any dimension, `schrodinger_of_preserves_vonNeumannEntropy` chooses the phases of the unitaries $U(t)$ so that $t \mapsto U(t)\psi$ is differentiable at t_0 for every ψ (Section 1.2). In finite dimension the same hypotheses make $t \mapsto U(t)$ differentiable in operator norm. In infinite dimension no choice of phases does. On $\ell^2(\mathbb{N})$ with basis (e_k) let

$$U(t) = 1 + (e^{it} - 1)\,|e_{k(t)}\rangle\langle e_{k(t)}|\,, \qquad k(t) = \lceil 1/|t| \rceil + 1,$$

which multiplies the component along $e_{k(t)}$ by e^{it} (`movingPhase`), and $\Phi_t(\rho) = U(t)\rho\,U(t)^\dagger$. At $t = 0$ the factor $e^{it} - 1$ vanishes, so $U(0) = 1$.

Results.

Statement	Lean
$1 + (z-1)\,\|e\rangle\langle e\|$ is unitary for $\|z\| = 1$	`phaseOp_mem_unitary`
$\|U\rho U^\dagger - \rho\| \le (2\|z-1\| + \|z-1\|^2)\,\|\rho e\|$ for self-adjoint ρ	`norm_phaseOp_mul_mul_star_sub_le`
$\|\rho e_n\| \to 0$ for a density operator ρ	`tendsto_norm_apply_hilbertBasis`
EVDIFF at 0 in operator norm, with derivative 0	`hasDerivAt_movingPhase_conj`
No unitary family inducing Φ_t near 0 is norm-differentiable at 0	`not_exists_differentiableAt_movingPhase`
All hypotheses of `schrodinger_of_preserves_vonNeumannEntropy` hold, and no norm-differentiable $U(t)$ exists	`movingPhase_counterexample`

Proof.

- Φ_t is conjugation by a unitary, so it maps density operators to density operators, preserves the entropy and equal mixtures, and $\sigma = U(t)^\dagger \rho\, U(t)$ is a reconstructed past.
- EVDIFF at 0: with $P = |e_k\rangle\langle e_k|$ and $a = e^{it} - 1$, $\Phi_t(\rho) - \rho = a\,P\rho + \bar{a}\,\rho P + |a|^2 P\rho P$, and $\|P\rho\| = \|\rho P\| = \|\rho e_k\|$. So $\|\Phi_t(\rho) - \rho\| \le (2|a| + |a|^2)\|\rho e_{k(t)}\|$, where $|a|/|t| \to 1$ and $\|\rho e_{k(t)}\| \to 0$ as $t \to 0$: by $\|\rho e_n\| \le \|\sqrt{\rho}\|\,\|\sqrt{\rho}\,e_n\|$ and $\sum_n \|\sqrt{\rho}\,e_n\|^2 = \operatorname{tr}\rho = 1$. The derivative is 0 in operator norm, for every density operator.

- No phase choice: if unitaries $V(t)$ induce Φ_t on pure states near 0, then $V(t) = c(t)U(t)$ with $c(t) = \langle e_0|V(t)e_0\rangle$ and $|c(t)| = 1$, since $U(t)e_0 = e_0$. If V is norm-differentiable at 0, so are c and $U = c^{-1}V$. For each n, $U(t)e_n = e_n$ near $t = 0$, so the derivative $U'(0)$ vanishes on every e_n, hence $U'(0) = 0$. But $\|U(t) - 1\| \geq \|(U(t) - 1)e_{k(t)}\| = |e^{it} - 1|$, and $|e^{it} - 1|/|t| \to 1$, so $U(t) - 1$ is not $o(t)$.

Notes.

- The equation still holds on vectors: $U(t)\psi$ has derivative 0 at 0 for every ψ, so $H = 0$ in `schrodinger_of_preserves_vonNeumannEntropy`. What fails is only the operator-norm form $U'(t_0) = (i\hbar)^{-1}HU(t_0)$, which the finite-dimensional `schrodinger_of_preserves` gives.
- EVDIFF holds here only at $t_0 = 0$: $k(t)$ jumps at $t = \pm 1/n$, and so does $U(t)$. A smooth variant (not formalized) puts the phase $\theta_k(t) = \varphi(kt)/k$, for a smooth bump φ supported in $[1, 2]$ with $\varphi(3/2) \neq 0$, on every e_k at once. Near each $t \neq 0$ only finitely many θ_k are nonzero, so $U(t)$ is norm-differentiable there; at 0, $\|\Phi_t(\rho) - \rho\| \leq 2\,\|U(t) - 1\|\,\|\rho Q_t\|$ with Q_t the projection onto $\overline{\operatorname{span}}\{e_k : k \geq 1/|t|\}$, and $\|\rho Q_t\| \to 0$ since ρ is compact. So EVDIFF holds at every time, also for the two-parameter family $U(t_1)U(t_0)^\dagger$, while $\|U(t)-1\| \geq |e^{i\varphi(3/2)/k}-1|$ at $t = 3/(2k)$, which is again not $o(t)$.
- With EVTI the propagator is $e^{t(i\hbar)^{-1}H}$ with bounded H, which is norm-differentiable; the example needs a time-dependent evolution.

Problems found in the brief.

- Step 3 differentiates $U(t)$ as an operator. In infinite dimension EVDIFF, even in operator norm on every density operator, does not give a norm-differentiable $U(t)$, only a strongly differentiable one; Step 3 then goes through on vectors, as in Section 1.2.

1.5 Unbounded Hamiltonians

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/Unbounded.lean`
- Imports: `SchroedingerEvolution.InfiniteDimensional`, the any-dimension part of the brief "Multiple equivalent characterizations of the Schrödinger equation" (for `IsEvolution`, `DrInfo`, `DrSceq` and `schrodinger_tfae`). That module imports `UnitaryEntropy.InfiniteDimensional`, so the module graph stays acyclic, but the library `UnitaryEntropy` now depends on the library `SchroedingerEvolution`.

Setting. As in Section 1.2, restricted to a time-independent evolution (EVTI): Φ_t for $t \geq 0$, with $\Phi_0 = I$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$, affine on the density operators and continuous at 0^+ in operator norm on each density operator (`IsEvolution`). DR-INFO is entropy preservation together with a reconstructed past for every density operator (`DrInfo`). DR-SCEQ is conjugation by a propagator U with $U(0) = I$ that solves $\frac{d}{dt}U(t)\psi = (i\hbar)^{-1}HU(t)\psi$ for ψ in the domain of a self-adjoint, in general unbounded, H (`DrSceq`). EVDIFF is weakened to a dense set of states (`EvDiff`): there is a dense subspace D such that $t \mapsto \Phi_t(|\psi\rangle\langle\psi|)$ is differentiable on $t \geq 0$, in operator norm, for every unit vector $\psi \in D$.

Results.

Brief	Lean
EVDIFF on a dense set of states	`EvDiff`
Theorem 2 ($\Rightarrow$), EVDIFF	`evDiff_of_drSceq` (with `differentiableAt_proj_of_hasDerivAt`)
Theorem 2 with EVTI, unbounded H	`drSceq_iff_drInfo_and_evDiff`

Differences from the brief.

- Only the time-independent case is formalized. For ($\Leftarrow$), DR-INFO gives DR-SCEQ by `schrodinger_tfae`, through DR-PROB, DR-EV, DR-UNIT and Stone's theorem; EVDIFF is not used, since continuity and the group law already give a self-adjoint generator. For ($\Rightarrow$), DR-INFO is part of `schrodinger_tfae`, and the dense subspace in EVDIFF is the domain of H: for ψ there, $t \mapsto U(t)\psi$ is differentiable, hence so is $t \mapsto |U(t)\psi\rangle\langle U(t)\psi| = \Phi_t(|\psi\rangle\langle\psi|)$, as $(x, y) \mapsto |x\rangle\langle y|$ is a bounded real-bilinear map.
- The time-dependent case of Theorem 2 with unbounded $H(t)$ is not formalized; see the next paragraph.

Problems found in the brief.

- For unbounded generators, Step 3 gives only a symmetric $H(t_0)$. If $t \mapsto U(t)\psi$ is differentiable at t_0 only for ψ in a dense subspace D, differentiating $\langle U(t)\psi|U(t)\phi\rangle = \langle\psi|\phi\rangle$ gives $\langle A\psi|\phi\rangle = -\langle\psi|A\phi\rangle$ on D: $A(t_0)$ is skew-symmetric, which is weaker than $A(t_0) = -A(t_0)^\dagger$ for an unbounded $A(t_0)$. Then $H(t_0)$ need not be self-adjoint, even on the largest such D. Example (not formalized): on $L^2(0,1)$ let H_α and H_β be $-i\,d/dx$ with the boundary conditions $\psi(1) = e^{i\alpha}\psi(0)$ and $\psi(1) = e^{i\beta}\psi(0)$, $\alpha \neq \beta$, both self-adjoint. Take $H(t) = H_\beta$ for $t < 0$ and $H(t) = H_\alpha$ for $t \geq 0$, so the family differentiated at $t_0 = 0$ is $U(t) = U_{0\to t} = e^{-itH_\alpha/\hbar}$ for $t \geq 0$ and $U(t) = U_{t\to 0}^\dagger = e^{-itH_\beta/\hbar}$ for $t \leq 0$. The evolution is unitary, so it satisfies [B]EVMIX and DR-INFO. The right derivative of $U(t)\psi$ at 0 exists exactly for ψ in the domain of H_α, the left one for ψ in the domain of H_β, so $U(t)\psi$ is differentiable at 0 exactly when $\psi \in H^1(0,1)$ with $\psi(0) = \psi(1) = 0$. This subspace is dense, but $-i\,d/dx$ on it is the minimal momentum operator, closed and symmetric with deficiency indices $(1,1)$, and not self-adjoint. So the self-adjointness of $H(t)$ has to be assumed, as in Kato's theory of time-dependent Hamiltonians, rather than derived from EVDIFF. With EVTI it is derived, by Stone's theorem.